

Machine Learning Applications for Health

健康领域的机器学习应用

COMP90089 (2026) - Lecture 1

COMP90089 (2026) - 第 1 讲

Prof Daniel Capurro Daniel Capurro 教授

About COMP90089 - Staff

关于 COMP90089——教职员工

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About COMP90089 - Tutors

关于 COMP90089——助教

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Learning Outcomes 学习成果

On completion of this subject, students should be able to:

完成本课程后，学生应能够：

- Describe the application of Machine Learning concepts in the context of health problems

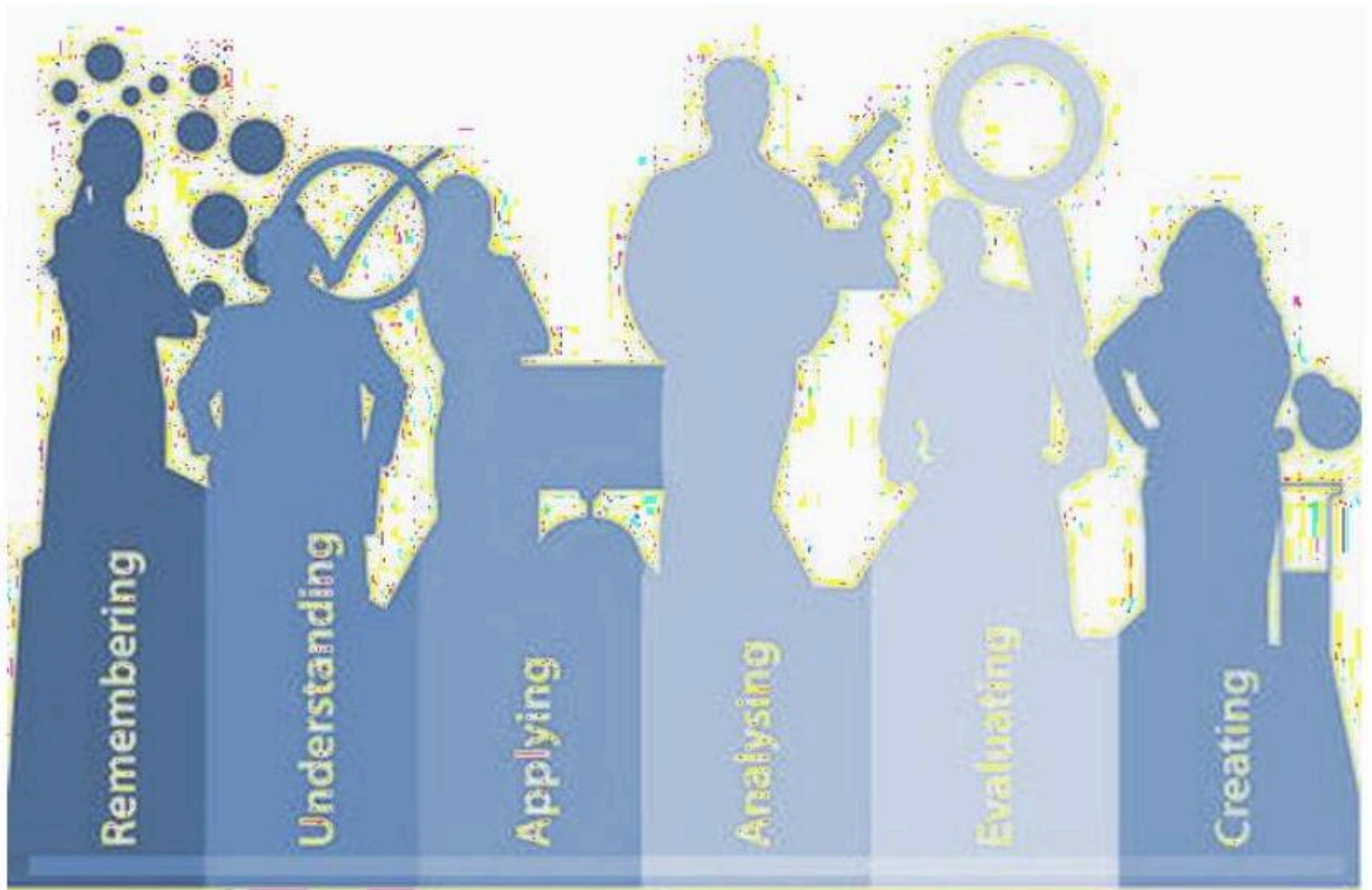
-描述机器学习概念在健康问题情境中的应用

- Demonstrate familiarity with challenges associated with health data and data modelling

-展示对健康数据及数据建模相关挑战的熟悉程度

- Design, implement, and evaluate a Machine Learning system addressing a healthcare problem

-设计、实现并评估一个解决医疗保健问题的机器学习系统



- Critically assess Machine Learning applications for health

-批判性评估机器学习在医疗保健领域的应用

About COMP90089 - Assessment

关于 COMP90089——考核

Regular (in class) quizzes worth 20% in total

定期（课堂内）测验，总计占 20%

- 3-4 questions covering the week's content

-涵盖本周内容的 3 至 4 道题

Two programming assignments worth 30% in total

两项编程作业，总计占 30%

- Assignment 1-15% -作业 1-15%
- Assignment 2-15% -作业 2——15%
 - o 50% hurdle collectively o总计 50% 的及格门槛
- Assignments/exam will require some programming

-作业/考试将要求进行一些编程

- o A mixture of SQL, Python, pseudocode and “explain with your own words”

oSQL、Python、伪代码以及“用你自己的话解释”的混合形式

Group Project worth 10% (proposal) + 40% (final report)

小组项目占比 10%（提案）+ 40%（最终报告）

- 50% hurdle -50%及格线
- 4-5 students in a group by week 3 - group contract to be signed

-第 3 周前每组 4 至 5 名学生——需签署小组契约

Organisations 组织

This subject is offered on campus only

本课程仅在校内授课

- The lectures are on campus

-讲座在校内进行

- Tutorials are on campus -辅导课在校内进行

- All lectures will be recorded. All recordings and other materials will be made available online through Canvas

-所有讲座都会录制。所有录音及其他材料都将通过 Canvas 在线提供

- Attending the lectures and tutorials is highly recommended.

-强烈建议参加讲座和辅导课。

Tutorials start on Week 2

辅导课从第 2 周开始

Quizzes start on week 2

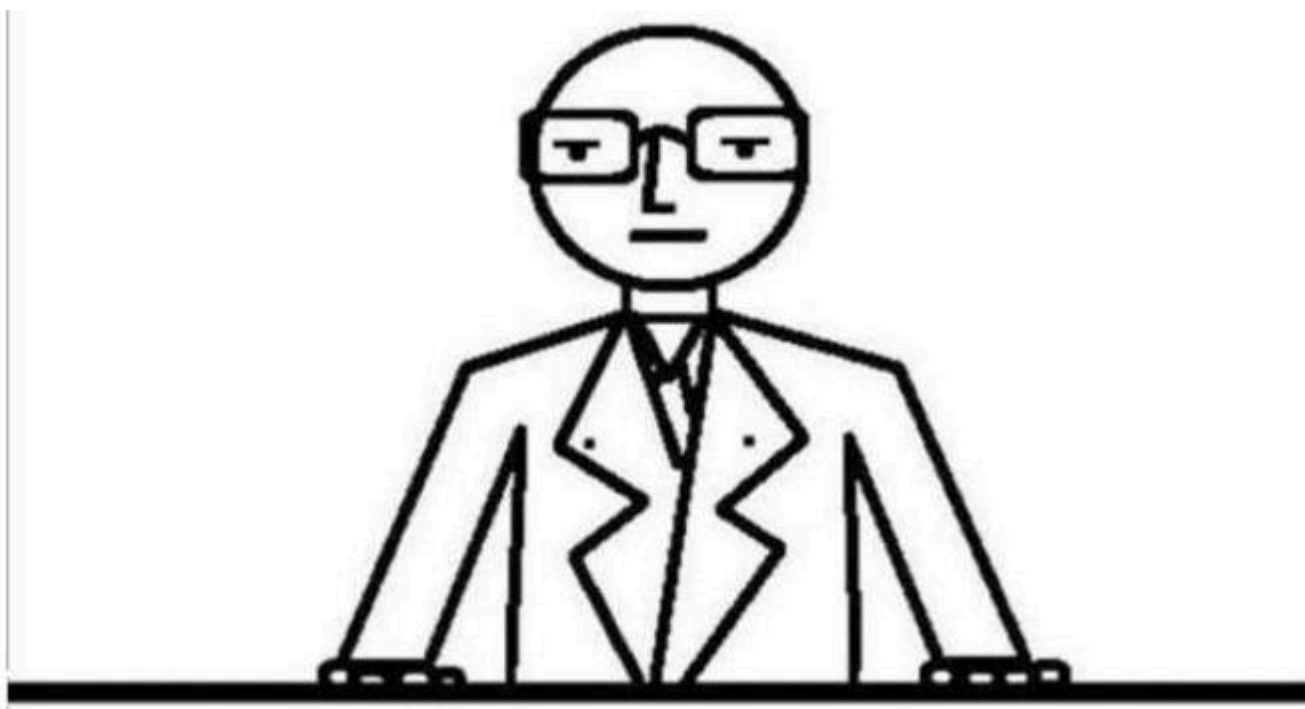
测验从第 2 周开始

Let's make this interactive

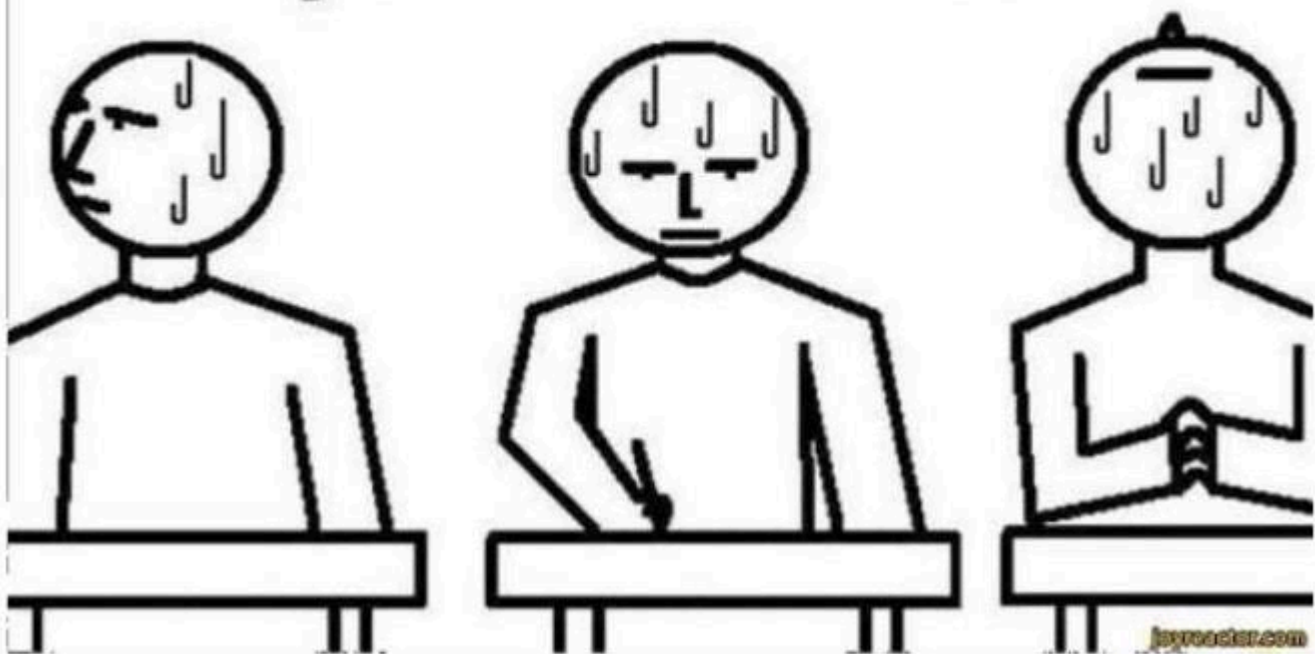
让我们进行互动吧

The teacher poses a question and scans the audience...

老师提出一个问题，扫视着听众.....



Must... Not... Make... Eye... Contact...



	Monday 星期一	Tuesday 星期二	Wednesday 星期三
DA:00			
09:00			09:00 12× w31-39, 41-43 COMP90089_U_1_SM2_2026_T01_04 Tutorial 1 09:00-10:00 PAR-192-L2-204-Flexible Learning Space (30)
10:00			
11:00			
12.00			
1300			13.00 12× w31-39, 41-43 COMP90089_U_1_SM2_2026_T01_04 Tutorial 1 13:00-14:00 PAR-115-L7-719-Collaborative Learning Space
12× w31-39, 41-43 12× w31-39, 41-43 COMP90089_U_1_SM2_2026_T01_04 COMP90089_U_1_SM2_2026_T01_04 Tutorial 1 Tutorial 1 14:00-15:00 14:00-15:00 PAP-104-L1-123-Flexible Learning Space (30) PAR-104-L1-123-Flexible Learning Space (30) 15.00	1400	15.00 12× w31-39, 41-43 COMP90089_U_1_SM2_2026_L01_01 Lecture 1 15:00-16:00 PAR-253-L2-230-Q230 Theatre (196) 16 : 00 – 16 : 00	14.00 – 14.00 14.00 12× w31-39, 41-43 COMP90089_U_1_SM2_2026_T01_04 Tutorial 1 14:00-15:00 PAR-115-L7-719-Collaborative Learning Space 15.00 15.00
12× w31-39, 41-43 12× w31-39, 41-43 COMP90089_U_1_SM2_2026_T01_04 COMP90089_U_1_SM2_2026_T01_04 教程 1 教程 1 14:00-15:00 14:00-15:00 PAP-104-L1-123-灵活学习空间 (30) PAR-104-L1-123-灵活学习空间 (30) 15.00		15.00 12× w31-39, 41-43 COMP90089_U_1_SM2_2026_L01_01 COMP90089_U_1_SM2_2026_L01_01 讲座 1 15:00-16:00 PAR-253-L2-230-Q230 剧场 (196) 16 : 00 – 16 : 00	14.00 – 14.00 12× w31-39, 41-43 COMP90089_U_1_SM2_2026_T01_04 教程 1 14:00-15:00 PAR-115-L7-719-协作学习空间
16.00		12× w31-39, 41-43 COMP90089_U_1_SM2_2026_L02_01 Lecture 2 16:00-17:00 PAR-253-L2-230-Q230 Theatre (196)	16.00
		12× w31-39, 41-43 COMP90089_U_1_SM2_2026_L02_01 第二讲 16:00-17:00 PAR-253-L2-230-Q230 阶梯教室 (196)	
17.00			

COMP90089 - Subject Communication

COMP90089 - 课程沟通

Ed Discussion

- Actively engage by asking and answering questions. Peer teaching is the most effective way of learning!

-积极参与，提出问题并回答问题。互相教学是最有效的学习方式！

- Please do not post questions too close to an assessment deadline

-请不要在考核截止日期临近时发布问题

- (Of course, no assignment solutions should be given away. Doing - or asking for - this is academic misconduct.)

-（当然，不得泄露任何作业解答。进行此类行为或提出此类要求均属于学术不端。）

Announcement 公告

- Subject updates, assignment announcement, schedule changes, etc.

-课程更新、作业公告、日程变更等。

Personal/private concerns: Email Daniel

个人/私人事宜：请发送电子邮件给 Daniel

- With private or personal concerns

-涉及私人或个人事宜

- Please include COMP90089 in the email subject

-请在邮件主题中注明 COMP90089

COMP90089 - Other Supports

COMP90089 - 其他支持渠道

More information on Canvas

Canvas 上的更多信息

- Tutorials location and time

-辅导课地点和时间

- Academic integrity -学术诚信

- University-wide support student services

-全校性学生支持服务

Independent Work and Plagiarism

独立作业与抄袭

Plagiarism and Misconduct

抄袭与学术不端

- The University takes plagiarism very seriously, whether deliberate or accidental. It is your responsibility to understand how to avoid plagiarism and collusion.

-无论是故意还是无意，大学都非常重视抄袭问题。你有责任了解如何避免抄袭和串通作弊。

- Details can be found in Canvas.

-详情请见 Canvas。

Academic honesty 学术诚信

- Submitted work are to be done independently.

-提交的作业必须独立完成。

- Please don't copy your work from anywhere else.

-请勿从其他任何地方抄袭作业。

- If cheating/collusion is detected, both parties - receiver and giver - will be referred to the

-如果发现作弊或串通行为，双方——接受者和提供者——都将被提交至

Faculty of Engineering for handling under the University Discipline procedures.

提交工程学院，按照大学纪律程序处理。

Student Representatives 学生代表

Student Representative 学生代表

- Would you like to represent the class on the student-staff liaison committee?

-你愿意代表全班参加师生联络委员会吗？

- You collect the concerns of your classmates and present them to the committee.

-收集同学们的意见，并将其提交给委员会。

- Contact me if you would like to volunteer.

-如果你愿意担任志愿者，请联系我。

FAQ - About the group project

常见问题——关于小组项目

- Can I pick my own topic for the project?

-我可以自行选择项目主题吗？

- Yes, but you need to agree as a group what your topic will be.

○可以，但你们需要以小组为单位就选题达成一致。

- Can I do my project individually?

-我可以单独完成项目吗？

- No ○不可以

- What can I do if my group is not working for me?

-如果我的小组不适合我，我该怎么办？

- Refer to the group contract you will sign by week 3 and please let the instructors know as soon as possible.

-请参阅你们将在第 3 周前签署的小组合同，并请尽快告知授课教师。

Feedback 反馈

Feedback 反馈

- Throughout the course -在整个课程期间
- At the end of the semester

-学期结束时

- Subject Experience Survey (SES)

○课程体验调查 (SES)

- Constructive criticisms are welcome

○欢迎提出建设性的批评意见

- Focus on things we can change & are under our control

○关注我们能够改变且可控的事项

Weekly quizzes 每周测验

- Review content and your progression

-复习课程内容并了解自己的学习进度

COMP90089 - What to expect?

COMP90089——课程预期

- Overview of Machine Learning flavours

-机器学习不同类型概览

- Paired with applications in health

◦结合医疗领域的应用

- Domain experts ◦领域专家

- No required ML/healthcare data experience

◦不要求具备机器学习或医疗数据经验

- Hands on experience with real clinical data

-使用真实临床数据的实践经验

- MIMIC critical care database

◦MIMIC 重症监护数据库

- Tutorials ◦辅导课

- Programming assignments ◦编程作业

- Group project ◦小组项目



Cogniti AI learning activities, your way

Cogniti AI 学习活动，随心打造

Cogniti is designed by educators, and built by educators, to empower educators.

Cogniti 由教育工作者设计并构建，旨在赋能教育工作者。

Create steerable and accurate agents powered by generative AI. Build mini apps where students can learn interactively. Set up sandboxes where students can build their own.

创建由生成式 AI 驱动、可控且准确的智能体。构建让学生能够互动学习的小型应用。搭建沙盒，让学生创建属于自己的作品。

Cogniti is designed to let teachers build activities that can be given specific instructions, and specific resources, to assist student learning in context-sensitive ways. Because teachers know how learning needs to work.

Cogniti 旨在让教师能够构建活动，为其设定具体指令并提供特定资源，以符合语境的方式辅助学生学习。因为教育工作者最了解学习应如何开展。



In-class Activity 课堂活动

Form a group with 2-6 classmates sitting near you and discuss the following:

与坐在你附近的 2—6 位同学组成小组，并讨论以下问题：

- What machine learning applications in healthcare have you seen or heard about before?

-你以前见过或听说过哪些医疗领域的机器学习应用？

- What machine learning applications for health would you love to create, something exciting or impactful?

-你最希望创建哪些用于健康领域的机器学习应用？可以是令人兴奋或具有重大影响的应用

Unique Challenges in Health Data

健康数据面临的独特挑战

- While immensely valuable, health data presents several inherent challenges that distinguish it from other data science domains.

-健康数据具有极高的价值，但也存在诸多固有挑战，使其有别于其他数据科学领域。

- Addressing these requires specialized techniques and careful consideration

-应对这些挑战需要专门的技术和审慎的考量

- ✓ Missing Data: Up to 30% in typical EHRs, requiring robust imputation strategies.

✓缺失数据：典型电子健康记录中的缺失数据比例最高可达 30%，需要采用稳健的插补策略。

- ✓ Systematic Biases: Inherited from data collection practices, leading to potential inequities in model performance across patient groups.

✓系统性偏差：源于数据收集实践，可能导致模型在不同患者群体中的表现存在不公平现象。

- ✓ High-Dimensional, Multi-Modal: Data comes in various formats (numerical, categorical, image, text, time-series), often with a very large number of features.

✓高维、多模态：数据以多种格式呈现（数值型、分类型、图像、文本、时间序列），通常包含数量极其庞大的特征。

- ✓ Strict Privacy Regulations: Laws like GDPR and HIPAA necessitate advanced privacy-preserving techniques and ethical data handling.

✓严格的隐私法规：GDPR 和 HIPAA 等法律要求采用先进的隐私保护技术，并以合乎伦理的方式处理数据。

- ✓ Interpretability Needs: Clinical decisions require ML models that are not just accurate but also explainable and trustworthy.

✓可解释性需求：临床决策要求机器学习模型不仅准确，还应具备可解释性和可信度。

- These unique challenges require specialized approaches beyond standard ML techniques.

-这些独特的挑战需要采用超越标准机器学习技术的专门方法。

Real-world data set: MIMIC

真实世界数据集：MIMIC

- Relational database -关系数据库
- Large-scale, freely available

-大规模、免费开放获取

- >40,000 patients stays in critical care units

○>40,000 名患者入住重症监护病房

- De-identified health-related database

○去标识化的健康相关数据库

- Beth Israel Deaconess Medical Center/US

○美国贝斯以色列女执事医疗中心

- Between 2001 and 2012 ○2001 年至 2012 年期间
- Highly granular data -高度细粒度的数据
 - Demographics ○人口统计信息
 - Vital signs ○生命体征
 - Lab results & procedures
 - 实验室检查结果与操作流程
 - Medications ○药物
 - Mortality ○死亡率



- Students are required to obtain access to the MIMIC-IV database

-学生必须获得 MIMIC-IV 数据库的访问权限

- Involves ethics training
 - 包括伦理培训
- A tutorial will be provided on Canvas
 - Canvas 上将提供教程

Python & SQL Python 和 SQL

- We'll use Python & SQL in our tutorials & assignments

-我们将在教程和作业中使用 Python 和 SQL

- Anaconda / Miniconda
 - Easy to manage libraries and dependencies
 - 易于管理库和依赖项



- Includes many commonly-used scientific libraries

◦包含许多常用的科学计算库



ANACONDA®

- Python packages ◦Python 软件包
 - <https://www.anaconda.com/download>
 - <https://conda.io/miniconda.html>



- Documentation: ◦文档:
 - <https://conda.io/docs/user-guide/>

Jupyter

Jupyter notebooks (<http://jupyter.org/>)

Jupyter notebooks (<http://jupyter.org/>)

- Service for Interactive Computing

-交互式计算服务

- Mix code, documentation, and plots, interactively

-交互式地混合使用代码、文档和图表

- Built for data science -专为数据科学而构建
- Great for reproducible analysis and exploration

-非常适合进行可复现的分析和探索

- We'll use this in our tutorials/assignments

-我们将在教程/作业中使用它

What do you need to do this week?

本周需要做什么？

- Sign up for a tutorial slot

-报名参加一个教程时段

- Get MIMIC-IV access -获取 MIMIC-IV 访问权限
 - o Google BigQuery
- Form your group -组建你们的小组
- Read course information on Canvas

-阅读 Canvas 上的课程信息

- Complete your pre-class activities before next week.

-在下周之前完成课前活动。

Sneak peek - Lecture 2

提前了解——第 2 讲



